
Diversity-Aware Recursive Feature Multiple Kernel Learning

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Abstract

Multiple kernel learning (MKL) combines several base kernels in the spirit of ensemble learning, yet existing methods rarely model kernel diversity—a known cornerstone of ensembles—and most traditional kernels weight all features uniformly, ignoring feature-level discriminability. We address both gaps with DARFMMKL: a data-driven kernel family (Recursive Feature Machine kernels) that learns feature importance directly from data, paired with a kernel selection method that jointly optimizes diversity and quality. The resulting NP-hard binary quadratic program is reformulated via Glover linearization and continuous relaxation into a linear program, and accelerated by Nyström sketching, yielding a selector whose cost is decoupled from the sample size. We provide a covering-number generalization bound that explicitly relates kernel diversity to estimation error. Experiments on 12 benchmark datasets show that DARFMMKL consistently outperforms 9 state-of-the-art MKL methods.

1. Introduction

Kernel methods (Schölkopf et al., 1999) have achieved significant success in various machine learning applications, including but not limited to classification, regression, ranking, clustering, and dimensionality reduction. The effectiveness of kernel methods is strongly influenced by the choice of kernel function, as the associated feature mapping has a profound impact on generalization performance. Consequently, the selection of an appropriate kernel plays an essential role in the success of kernel methods.

In the context of kernel learning, it is often more advantageous to select or combine multiple kernels rather

than exhaustively searching for a single optimal kernel (Gönen & Alpaydin, 2011), giving rise to *Multiple kernel learning* (MKL). Most existing MKL methods aim to learn a weighted linear (Lanckriet et al., 2004; Rakotomamonjy et al., 2008) or non-linear combination (Varma & Babu, 2009; Cortes et al., 2009) of kernels from a predefined set by constructing an optimization problem. Some approaches assign kernel weights based on different measures of kernel quality, such as kernel alignment (Cristianini et al., 2001; Kandola et al., 2002; Qiu & Lane, 2009; Cortes et al., 2010; Alavi & Hashemi, 2022), feature space-based kernel matrix evaluation measures (Nguyen & Ho, 2008), and KL divergence (Ying et al., 2009), while others formulate different structural risk optimization problems, including semi-definite programming or quadratically constrained quadratic programming with various restrictions on kernel properties (Do et al., 2009; Kloft et al., 2011; Liu et al., 2017; Han et al., 2018). The primary limitation of these MKL methods lies in their insufficient consideration of diversity among kernel functions, susceptibility to disturbance from suboptimal kernels, and poor scalability to large datasets.

Numerous studies (Kuncheva & Whitaker, 2003) have demonstrated the importance of diversity among classifiers in ensemble learning (Zhou, 2012). Xia & Hoi (2012) and Shen et al. (2021) introduced ensemble methods into kernel learning, indirectly highlighting the importance of kernel diversity, but neither has explicitly quantified it. In the field of multiple kernel clustering, the Frobenius product and squared Frobenius divergence between kernel matrices have been used to represent kernel diversity (Liu et al., 2016; 2020; Yao et al., 2020). However, the computational cost remains expensive due to the alternating optimization procedure and calculations involving kernel matrices.

To address the computational burden arising from large datasets, several methods have been proposed, including random Fourier feature (RFF) methods (Rahimi & Recht, 2007; 2008), Nyström methods (Williams & Seeger, 2000; Drineas & Mahoney, 2005), and coresets methods (Tsang et al., 2005; 2006). The RFF method approximates shift-invariant kernels with orthogonal trigonometric function families and is data-independent, which results in weaker generalization performance. The coreset approximation method and the Nyström method differ essentially in their

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